



Does temperature modify the effect of PM₁₀ on mortality? A systematic review and meta-analysis[☆]



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ABSTRACT

Large and growing literature has explored whether temperature modified the effect of particular matter (PM) on mortality, but results of the modification effect are inconsistent. In this study, we reviewed information from 29 studies to get the qualitative evidence of the modification effects of temperature on PM to mortality, and the data from 16 of the 29 studies were extracted to conduct a meta-analysis. Temperatures were grouped into three level: “low”, “middle” and “high” according to the original studies. The random effect model was used in the meta-analysis with the relative risk (RR) as the measure indicator. The RRs (95% confidence intervals, CIs) for non-accidental death, cardiovascular death and respiratory death per 10 µg/m³ increase in PM₁₀ were 1.004 (1.003, 1.006), 1.005 (1.003, 1.007), and 1.005 (1.000, 1.010) in the low temperature level, 1.005 (1.004, 1.006), 1.005 (1.004, 1.007), and 1.008 (1.006, 1.010) in the middle temperature level, and 1.012 (1.010, 1.015), 1.016 (1.010, 1.022) and 1.019 (1.010, 1.028) in the high temperature level, respectively. In conclusion, moderate evidence exists that temperature modifies the effect of PM₁₀ on mortality. The effect of PM₁₀ on respiratory death was the greatest, while the effect on non-accidental death was the smallest in the same temperature level. In addition, the effects of PM₁₀ on all the three kinds of mortality were the biggest in the high-temperature level, and the smallest in the low-temperature level.

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1. Introduction

The adverse effect of particular matter (PM) on human mortality have been established in a series of epidemiological studies around the world (Chen et al., 2013; Lee et al., 2014; Tong et al., 2012). Meanwhile, the effect of ambient temperature on mortality is also an important public health issue. Numerous studies have reported the independent effect of these two types of factors on mortality (Anderson and Bell, 2009; Basu, 2009; Ma et al., 2014; Wu et al., 2013). Recently, large and growing literature studied the effects of temperature modification on PM-associated mortality (Kim et al., 2015; Rodopoulou et al., 2015; Sun et al., 2015). Most of these studies identified significant interaction between PM and temperature, with stronger health effects of PM in high temperature days for mortality (Qian et al., 2008; Ren and Tong, 2006; Sun et al., 2015). However, other studies reported non-significant modification effect (Dholakia et al., 2014; Hales et al., 2000; Stafoggia et al.,

2008). Therefore, the findings of the effects of temperature modification on PM-associated mortality are not consistent. Moreover, to date, few reviews have specifically focused the effects of temperature modification on PM-associated mortality.

To systematically review the existing evidence of the effects of temperature modification on PM-associated mortality, we undertook a systematic review of original quantitative research. Firstly, we aimed to assess modification of temperature on the association between PM and cause-specific mortality. As a secondary aim, we also conducted meta-analyses to explore the magnitude of heterogeneity. In addition, the third aim was to describe the distribution of the current epidemiological studies and make recommendations for future research directions.

2. Methods and material

2.1. Search strategy and selection criteria

We systematically searched PubMed and Web of Science, which was limited to peer-reviewed journal articles published in English before June 12, 2016. Detailed search terms were: “air pollut*”,

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“particulate matter”; “meteorolog*”, “temperature”; “modif*”, “interact*”; “death”, “mortality” (see Fig. 1 for search results). On the one hand, the “and” was used between the four groups of keywords, the first group (the first two) indicating the PM, the second group (the second two) representing temperature, the third group (the third two) meaning modification effects and the fourth group (the last two) indicating mortality. On the other hand, the “or” was used to connect keywords within each group. The detail search process could be seen in the supplementary material Part A. Based on the preferred reporting items for systematic reviews and meta-analyses (PRISMA) statement guidelines (Moher et al., 2010), two reviewers (Fei Chen and Zhiwei Fan) conducted all the procedures independently and research team adjudicated any discrepancies. The objective literature was obtained with the steps as follows. First, we combined the keywords to search the literature from the two databases, and then eliminated those duplicated records. After that, reading the title and abstract, we screened out those articles without the specific aim of studying the interaction. Lastly, we read the remainder full-text articles in-depth to identify the target articles.

2.2. Study eligibility criteria and quality of the studies

We selected papers based on the following criteria. It must investigate the effects of temperature modification on PM-associated mortality. Moreover, original studies were included in the review if (1) there were temperature, PM and mortality in the original studies; (2) explicit statistical analysis on the modification effects; (3) clear conclusion of the modified effects. We limited the language only in English and included only peer-reviewed original articles. Those articles were excluded when (1) they focused on the

animal research; (2) the paper was a review or report.

We evaluated the basic characteristics and quality of the methodology of the studies included in the systematic review by extracting the following data: study area, study period, PM, specific mortality, main findings, statistical methods, modeling strategy, study design and sample size (Table 1). The two reviewers independently worked on data extraction, evaluation of study quality and classification of the evidence. The agreement was reached via consensus. Because of our review focusing on the association between air pollution and mortality, the most common used study design is observational design. To date, no validated scales have been developed to assess the quality of time-series and case-crossover studies. We evaluated the quality of the studies and obtained a quality level for each study, based on an adapted version of the criteria (see Supplemental material, Part B) used in a previous review (Li et al., 2016), which was developed according to New Castle Ottawa Scale, the Cochrane risk of bias tool and other instruments (Deeks et al., 2003; Higgins et al., 2011; Moher et al., 2010; Wells et al., 2000).

2.3. Meta-analysis

The original studies include in the systematic review, which employed stratification approach and classified temperature into different levels to explore the effects of temperature modification on PM₁₀-associated mortality, and gave RR or β value for PM₁₀, were chose to conduct the meta-analysis. We limited the meta-analyses to PM₁₀-associated mortality because studies focusing on PM_{2.5}-associated mortality was not enough to conduct the meta-analysis. We made the strategy to extract the point estimate and confidence interval of PM₁₀ regression coefficients for different mortality in

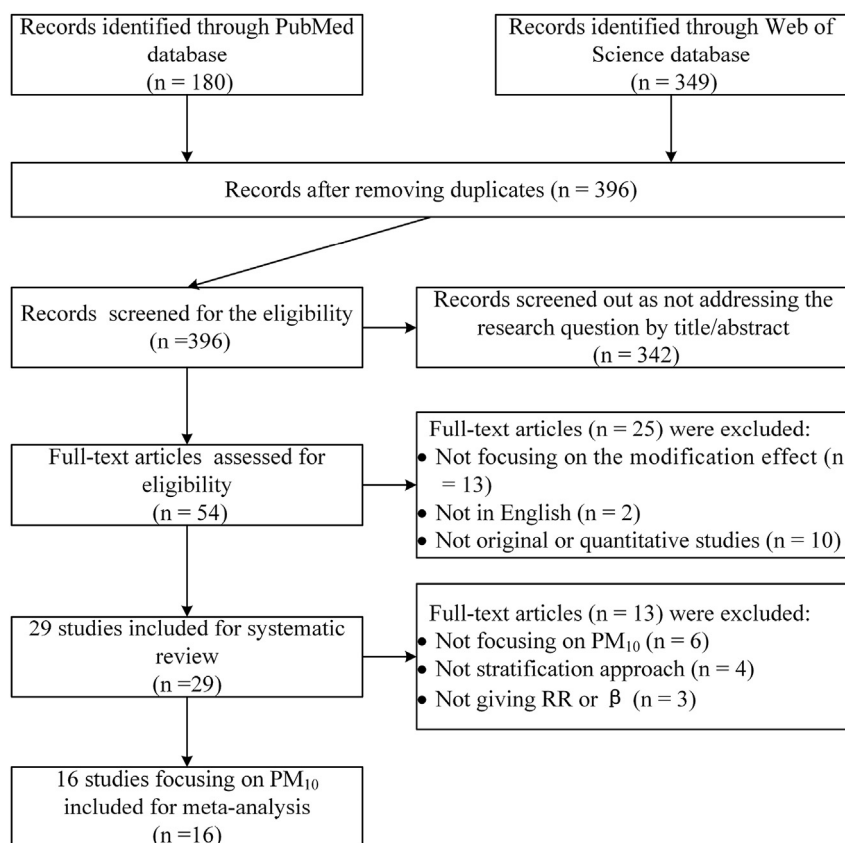


Fig. 1. Results of the systematic literature search.

Table 1
Characteristics of included studies.

Study	Study area	Study period	PM	Mortality	Main findings	Statistical methods	Study design and Modeling strategy	Study sample size
(Kioumourtoglou et al., 2016)	USA (207 cities)	2000–2010	PM _{2.5}	Congestive heart failure; Myocardial infarction; COPD; diabetes	A higher association between long-term PM _{2.5} exposure and mortality in warmer cities. Lag 0 for PM _{2.5} and temperature.	GLM;	Cohort; Interaction terms	11,411,282 deaths
(Wang et al., 2016)	New Jersey, USA	2004–2009	PM _{2.5}	Non-accidental deaths	Seasonal temperatures modify the effect of PM _{2.5} on mortality. Lag 0 for PM _{2.5} and temperature.	GLM	Time-series; Interaction terms	365,530 non-accidental deaths
(Sun et al., 2015)	Hong Kong, China	1999–2011	PM _{2.5}	Non-accidental deaths; Cardiovascular diseases; Respiratory diseases	The mortality effects of PM _{2.5} were stronger in low-temperature stratum than those in high. Lag 0–1 for PM _{2.5} and lag 0–1, 0–6, 7–13 for temperature.	GAM	Time-series; Stratification	456,317 non-accidental deaths; 125,031 cardiovascular disease deaths; 88,525 respiratory disease deaths.
(Li et al., 2015)	Beijing, China	2005–2009	PM _{2.5}	Non-accidental deaths; Cardiovascular diseases; Respiratory diseases	Temperatures could modify the effect of PM _{2.5} on mortality. Lag 0, 0–1, 0–2 for PM _{2.5} and lag 0 for temperature.	GAM	Time-series; BRS; Stratification	3861 respiratory deaths, 15,887 cardiovascular deaths, and 35,024 total non-accidental deaths
(Li et al., 2013a)	Beijing, China	2004–2009	PM _{2.5}	Respiratory diseases	The adverse effects of fine particles on respiratory illnesses were magnified in low-temperature. Lag from 0 to 8 for PM _{2.5} and lag 0 for temperature.	GAM	Time-series; Stratification	About 62 respiratory diseases deaths a day.
(Li et al., 2013b)	Beijing, China	2005–2009	PM _{2.5}	Non-accidental deaths; Cardiovascular diseases; Respiratory diseases	Modifying effect of PM _{2.5} by temperature exists in the study period. Lag from 0 to 10 for PM _{2.5} and lag 0 for temperature.	GAM	Time-series; BRS	The daily average 146 for non-accidental mortality, 70 for cardiovascular mortality and 16 for respiratory mortality.
(Willers et al., 2016)	Rotterdam, Dutch	1995–2009	PM ₁₀	Non-accidental deaths	It found significant synergistic effects of high temperatures and PM ₁₀ on mortality. Lag 0, 1, 2 for PM ₁₀ and temperature.	GLM	Case cross-over; Interaction terms	73,178 non-accidental deaths
(Kim et al., 2015)	South Korea (seven cities)	2000–2009	PM ₁₀	Non-accidental deaths; Cardiovascular diseases; Respiratory diseases	Temperature can affect the relationship between the PM ₁₀ levels and cause-specific mortality. Lag from 0 to 5 for PM ₁₀ and temperature.	GAM	Time-series; Stratification	828,787 non-accidental deaths, 230,218 cardiovascular deaths and 51,902 respiratory deaths.
(Shaposhnikov et al., 2014)	Moscow, Russia	2006–2010	PM ₁₀	Non-accidental deaths	Interaction between high temperatures and PM ₁₀ added a substantial number of deaths. Lag 0–1, 2–6 for PM ₁₀ and temperature.	GAM	Time-series; Stratification	About 300 deaths a day and non-accidental deaths accounting 95–96%
(Li et al., 2014)	Guangzhou, China	2005–2009	PM ₁₀	Non-accidental deaths; Cardiovascular diseases; Respiratory diseases; Cardiopulmonary diseases	PM ₁₀ effects were more adverse on high-temperature days. Lag 0, 1, 0–1 for PM ₁₀ and lag 0 for temperature.	GAM	Time-series; Stratification	41.2 nonaccidental deaths per day, 12.6 cardiovascular mortality per day, 7.8 respiratory mortality per day, and 20.4 cardiopulmonary mortality per day.
(Dholakia et al., 2014)	India (five cities)	2005–2012	PM ₁₀	Non-accidental deaths	No significant modification effect of temperature on PM ₁₀ -mortality association. Lag 1 for PM ₁₀ and lag 0 for temperature.	GAM	Time-series; Interaction terms; Stratification	The daily mean numbers of Non-accidental varied according to the size of the city and ranged from 4.2 to 225.6.
(Pinheiro et al., 2014)	Sao Paulo, Brazil	1998–2008	PM ₁₀	Cardiovascular diseases; Respiratory diseases	A higher relative risk for cardiovascular and respiratory mortalities at low temperatures and high temperatures. Lag 0 for PM ₁₀ and temperature.	GAM	Case cross-over; Interaction terms	The average of the daily death for cardiovascular and respiratory disease was 55.9 and 15.6.
(Puza and Roberts, 2013)	Australia (2 cities)	1993–2007	PM ₁₀	Non-accidental deaths	An interaction effect between PM and temperature in Sydney and no support for such an effect in Melbourne. Lag 0–2 for PM ₁₀ and lag 0 for temperature.	Bayesian model	Time-series; Stratification	No. of cases not reported
(Meng et al., 2012)	China (8 cities)	2001–2008	PM ₁₀	Non-accidental deaths; Cardiovascular diseases	Extreme high temperature increased the associations of PM ₁₀ with daily mortality. Lag 0–1 for PM ₁₀ and lag 0 for temperature.	GAM	Time-series; BRS; Stratification	The daily mean numbers of Non-accidental, cardiovascular and respiratory deaths varied according to the size of the city and ranged from

Table 1 (continued)

Study	Study area	Study period	PM	Mortality	Main findings	Statistical methods	Study design and Modeling strategy	Study sample size
(Cheng and Kan, 2012)	Shanghai, China	2001–2004	PM ₁₀	Respiratory diseases Non-accidental deaths; Cardiovascular diseases; Respiratory diseases	The acute health effects of air pollution might vary by temperature level. Lag 0 for PM ₁₀ and temperature.	GAM	Time-series; Stratification	11 to 119, from 7 to 44, and from 1 to 15, respectively 119 non-accidental mortalities per day, 44.2 cardiovascular mortality per day, 14.3 respiratory mortality per day.
(Park et al., 2011)	Seoul, South Korea	1999–2007	PM ₁₀	Non-accidental deaths; Cardiovascular diseases; Respiratory diseases	Temperature strongly modified the adverse effect of air pollution. Lag 0–1 for PM ₁₀ and temperature.	GLM	Time-series; Stratification	The daily average ranged from 55 to 149 for non-accidental mortality, 8–45 for cardiovascular mortality and 0–21 for respiratory mortality.
(Li et al., 2011)	Tianjin, China	2007–2009	PM ₁₀	Non-accidental deaths; Cardiovascular diseases; Respiratory diseases; Cardiopulmonary diseases	The interactions between PM ₁₀ and temperature were statistically significant on cardiovascular, cardiopulmonary, and ischemic heart diseases mortalities. Lag 0–1 for PM ₁₀ and lag 0 for temperature.	GAM	Time-series; BRS; Stratification	101 non-accidental mortalities per day, 53 cardiovascular mortality per day, 9 respiratory mortality per day, 62 cardiopulmonary mortality per day.
(Lin and Liao, 2009)	Kaohsiung, China	1995–1999	PM ₁₀	Non-accidental deaths; Cardiovascular diseases	Higher PM ₁₀ effects on mortality during high-temperature days with no statistically significant for non-accidental mortality. Lag 0, 1, 2, 3–5 for PM ₁₀ and temperature.	GAM	Time-series; Stratification	33,818 non-accidental deaths
(Stafoggia et al., 2008)	Italy (9 cities)	1997–2004	PM ₁₀	Non-accidental deaths; Cardiovascular diseases; Respiratory diseases	Higher PM ₁₀ effects on mortality during high-temperature days. Lag 0–1 for PM ₁₀ and temperature.	GLM	Case cross-over; Stratification	321,024 non-accidental deaths; 132,262 cardiovascular disease deaths; 22,151 respiratory disease deaths.
(Qian et al., 2008)	Wuhan, China	2001–2004	PM ₁₀	Non-accidental deaths; Cardiovascular diseases; Respiratory diseases; Cardiopulmonary diseases	Synergistic effects of PM ₁₀ and high temperatures on daily mortality. Lag 0–1 for PM ₁₀ and lag 0 for temperature.	GAM	Time-series; Stratification	89,131 non-accidental deaths, with a main contribution of cardiopulmonary mortality (daily mean of 35)
(Nawrot et al., 2007)	Flanders, Belgium	1997–2003	PM ₁₀	Non-accidental deaths; Cardiovascular diseases; Respiratory diseases	Mortality was highest in the first quintile (<7 °C) and lowest in the fourth quintile (14–17 °C). Lag 0–2, 0–3, 0–5 for PM ₁₀ and lag 0 for temperature.	GAM	Time-series; Stratification	354,357 non-accidental deaths;
(Ren and Tong, 2006)	Brisbane, Australia	1996–2001	PM ₁₀	Non-accidental deaths; Cardiovascular diseases;	PM ₁₀ exhibited more adverse health effects on warm days than cold days. Lag 0, 1, 2 for PM ₁₀ and temperature.	GAM GLM	Time-series; BRS; Interaction terms; Stratification	The average of the daily death for non-accidental and cardiovascular disease was 15.86 and 7.26.
(Roberts, 2004)	Cook, Allegheny, USA	1987–1994	PM ₁₀	Non-accidental deaths	The most consistent finding was that PM ₁₀ has the largest effect on mortality on hot days. Lag 0, 1, 0–1, 0–3 for PM ₁₀ and lag 0, 0–3 for temperature.	GAM	Time-series; BRS; Stratification	The average of the daily death for Cook and Allegheny County was 83 and 30.
(Aga et al., 2003)	Europe (28 cities)	About five years	PM ₁₀	Non-accidental deaths	The effect size was modified by temperature. Lag 0–1 for PM ₁₀ and lag 0 for temperature.	GAM	Time-series; Stratification	No. of cases not reported
(Katsouyanni et al., 2001)	Europe (29 cities)	About five years	PM ₁₀	Non-accidental deaths	Larger effect estimates found during the high temperature. Lag 0–1 for PM ₁₀ and lag 0, 1, 2, 0–2 for temperature.	GAM	Time-series; Stratification	The annual all-cause mortality rate per 100,000 ranged from 579 to 1231
(Burkart et al., 2013)	Berlin, German Lisbon, Portugal	1998–2010 1998–2008	PM ₁₀	Non-accidental deaths	Interaction effects between PM ₁₀ and high temperatures were mostly observed for Lisbon while such interactions were less obvious for Berlin. Lag 0–1, 0–13 for PM ₁₀ and temperature.	GAM	Time-series; BRS; Stratification	425,157 deaths in Berlin, 273,429 deaths in Lisbon,
(Muggeo, 2007)	Palermo, Italy		PM ₁₀					No. of cases not reported

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Table 1 (continued)

Study	Study area	Study PM period	Mortality	Main findings	Statistical methods	Study design and Modeling strategy	Study sample size
		1997–2001	Non-accidental deaths	The effect of PM ₁₀ on mortality is temperature-dependent. Lag 0, 0–15 for PM ₁₀ and temperature.	GAM DLM	Time-series; BRS	
(Hales et al., 2000)	Christchurch, New Zealand	1988–1993	PM ₁₀ Non-accidental deaths; Cardiovascular diseases; Respiratory diseases	No evidence of interaction between the effects of temperature and particulate. Lag 1 for PM ₁₀ and temperature.	GLM	Time-series; Interaction terms	The daily average of non-accidental, cardiovascular and respiratory mortality was 7.2, 2.3 and 0.8
(Pascal et al., 2014)	French (9 cities)	2000–2006	PM _{2.5} PM ₁₀ Non-accidental deaths; Cardiovascular diseases; Respiratory diseases; Cardiac disease; Ischemic disease; Cerebrovascular disease	A significant interaction between warm days and PM on mortality. Lag 0–1 for PM ₁₀ and lag 0 and 1–7 for temperature.	GAM	Time-series; Stratification	A total of 548,474 deaths

different temperature level. Temperatures of the original studies were grouped into three level: “low”, “middle” and “high”. If papers gave more than three levels, we treated the lowest level as “low”, the highest level as “high” and others as “middle”. Meanwhile, for studies reporting only two categories, we considered the lower one as “low” and the other as “high”. We contacted authors for additional data for our meta-analysis where there were not presented. Because of the small number of studies included in each meta-analysis, random-effects models were used to synthesize the modified effects (Lau et al., 1997). RR and its corresponding 95% confidence intervals (95% CIs) was used to assess the influence of PM₁₀ on mortality. The summary estimates were weighted by the inverse variance of each study. Forest plots were used to summarize the results of included studies. We described the between-study heterogeneity by using the I^2 metric. We assessed publication bias using the Egger's test for asymmetry (Egger et al., 1997). For reporting, we followed the meta-analysis of observational studies in epidemiology and the preferred reporting items for systematic review and meta-analysis guidelines (Moher et al., 2010; Stroup et al., 2000). In addition, to quantify the differences of RRs of associations between PM₁₀ and mortality, we adopted the difference of RRs between different temperature levels. Both point estimates and 95% confidence intervals were calculated using the formula shown below:

$$(\hat{Q}_1 - \hat{Q}_2) \pm 1.96 \left(\sqrt{(\widehat{SE}_1)^2 + (\widehat{SE}_2)^2} \right)$$

where $\hat{Q}_1$, $\hat{Q}_2$ are the estimates of RR for the two categories, and $\widehat{SE}_1$, $\widehat{SE}_2$ are their respective stand errors.

2.4. Sensitive analysis and subgroup analysis

Sensitivity analyses were performed by excluding the studies with low quality in the twenty-nine studies. We employed subgroup analyses to explore the sources of heterogeneity. Analyses were performed using R software with “metafor” package (V.3.31 <https://cran.r-project.org/>).

3. Results and discussion

3.1. Study selection

Fig. 1 shows the procedure of selecting literature. 396 studies were identified from electronic databases after removing the duplication. Based on the title and the abstract, 342 papers were excluded, with 54 full-text articles to be retrieved and assessed for eligibility eventually. Among these, 25 articles were excluded for the following reasons. 13 papers did not directly focus on the modification effects. Two records were not written in English. 10 was not original or quantitative studies. The remaining 29 papers were considered to be eligible for this systematic review. Then, six studies were excluded in the meta-analysis for they focusing on PM_{2.5}; three studies were eliminated for they did not use stratification approach; four studies were removed for they did not provide RR or β value. At last, 16 of the 29 papers were chosen to conduct the meta-analysis.

3.2. Study characteristics

According to our strategy, we identified 29 papers of original research directly related to the effects of temperature modification on PM-associated mortality. Similar studies are increasing rapidly, though these papers can help raise awareness of the effects of temperature modification on PM-associated mortality, and identify promising strategies for future evaluation, more systematic review are now needed to summarize existing results, and consequently identify new directions. 16 studies from the 29 eligibility studies, which quantified modification effects on PM₁₀-associated mortality by dividing temperature into several levels, were chose to conduct the meta-analysis.

Table 1 shows the detailed information of the studies included in the review. Among the eligible studies, there were six focusing on the effects of temperature modification on PM_{2.5}-mortality (Kioumourtoglou et al., 2016; Li et al., 2013a, 2013b, 2015; Sun et al., 2015; Wang et al., 2016); 22 studies explored the effects of temperature modification on PM₁₀-mortality, and one study focused on both PM_{2.5} and PM₁₀ (Pascal et al., 2014). The numbers of studies focusing on non-accidental mortality, cardiovascular diseases mortality and respiratory diseases mortality were 26, 17 and 16 respectively. Other specific mortality evaluated were

congestive heart failure, myocardial infarction, chronic obstructive pulmonary disease, diabetes, cardiopulmonary diseases, ischemic disease and cerebrovascular disease (Kioumourtzoglou et al., 2016; Pascal et al., 2014). Fig. 2 shows the study cities of all the 29 studies. Ten studies were conducted in China (Cheng and Kan, 2012; Li et al., 2011, 2013a, 2013b, 2014, 2015; Lin and Liao, 2009; Meng et al., 2012; Qian et al., 2008; Sun et al., 2015), three in USA (Kioumourtzoglou et al., 2016; Roberts, 2004; Wang et al., 2016), three in French (Aga et al., 2003; Katsouyanni et al., 2001; Pascal et al., 2014), four in Italy (Aga et al., 2003; Katsouyanni et al., 2001; Muggeo, 2007; Stafoggia et al., 2008), and the other 11 studies distributed in other countries. 25 of the included studies were time series studies; three were case cross-over studies (Pinheiro et al., 2014; Stafoggia et al., 2008; Willers et al., 2016) and only one was cohort study (Kioumourtzoglou et al., 2016). The most commonly used statistical model was GAM and GLM, with only one study using Bayesian model (Puza and Roberts, 2013). Interaction terms, stratification and bivariate response surface (BRS) were the three main ways to study the modification effects. Overall, from the main finding of these studies, the effects of temperature modification on PM-mortality was statistically significantly, but some studies still reported non-significant modification effect (Dholakia et al., 2014; Hales et al., 2000; Stafoggia et al., 2008). In addition, many studies estimated the modified effects on various lag days, and we chose lags in day 0 and 1 (lag0, lag1 or lag0-1).

3.3. Quality of included studies

The quality of twenty-nine studies were evaluated (see Supplemental material, Table 1). Five studies were classified as high quality (Muggeo, 2007; Park et al., 2011; Ren and Tong, 2006; Stafoggia et al., 2008; Sun et al., 2015); Twenty studies were classified as intermediate quality; The other four were classified as low quality for they did not report whether causes of mortality were coded according to the International Classification of Disease (Burkart et al., 2013; Dholakia et al., 2014; Puza and Roberts, 2013; Roberts, 2004). Overall, the quality of these studies was good with high and intermediate quality accounting for 86.2%.

3.4. The effects of temperature modification on PM-associated mortality

Twenty-two studies investigated effects of temperature modification on PM₁₀-associated nonaccidental mortality. Seventeen reported the effect size of PM₁₀ on non-accidental mortality was modified by temperature (Aga et al., 2003; Cheng and Kan, 2012; Katsouyanni et al., 2001; Kim et al., 2015; Li et al., 2011, 2014; Meng et al., 2012; Muggeo, 2007; Nawrot et al., 2007; Park et al., 2011; Pascal et al., 2014; Qian et al., 2008; Ren and Tong, 2006; Roberts, 2004; Shaposhnikov et al., 2014; Stafoggia et al., 2008; Willers et al., 2016); Two multi-city studies detected contradictory results (Burkart et al., 2013; Puza and Roberts, 2013); Three found no statistically significant interaction (Dholakia et al., 2014; Hales et al., 2000; Lin and Liao, 2009). Fourteen studies examined effects of temperature modification on PM₁₀-associated cardiovascular mortality (Cheng and Kan, 2012; Hales et al., 2000; Kim et al., 2015; Li et al., 2011, 2014; Lin and Liao, 2009; Meng et al., 2012; Nawrot et al., 2007; Park et al., 2011; Pascal et al., 2014; Pinheiro et al., 2014; Qian et al., 2008; Ren and Tong, 2006; Stafoggia et al., 2008). Eleven found the effect size of PM₁₀ on cardiovascular mortality was modified by temperature, while the other three found no interaction (Hales et al., 2000; Li et al., 2014; Lin and Liao, 2009). In addition, twelve studies examined effects of temperature modification on PM₁₀-associated respiratory mortality (Cheng and Kan, 2012; Hales et al., 2000; Kim et al., 2015; Li et al., 2011, 2014; Meng et al., 2012; Nawrot et al., 2007; Park et al., 2011; Pascal et al., 2014; Pinheiro et al., 2014; Qian et al., 2008; Stafoggia et al., 2008). Eleven found the effect size of PM₁₀ on cardiovascular mortality was modified by temperature, while the other one found no interaction (Hales et al., 2000). Moreover, seven studies investigated effects of temperature modification on PM_{2.5}-associated mortality, and all reported the effect size of PM_{2.5} on mortality was modified by temperature (Kioumourtzoglou et al., 2016; Li et al., 2013a, 2013b, 2015; Pascal et al., 2014; Sun et al., 2015; Willers et al., 2016). The present systematic review showed evidence that temperature modifies the effects of PM on mortality.

It is biologically plausible that temperature could modify the effects of PM on mortality. For PM₁₀, most studies found the effect

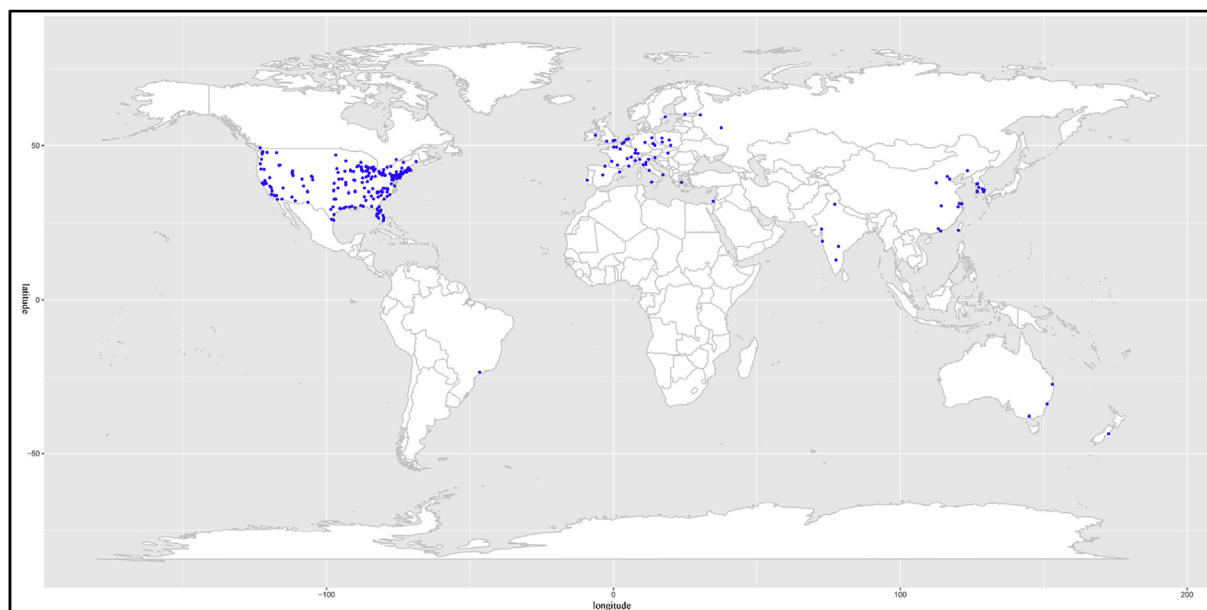


Fig. 2. The study cities of the 29 studies during the study period.

of PM₁₀ on mortality was larger during hot days (Kim et al., 2015; Li et al., 2011; Lu et al., 2015; Meng et al., 2012; Park et al., 2011; Pascal et al., 2014). For PM_{2.5}, though all the seven studies found the effects of temperature modification on PM_{2.5}-associated mortality, the modification effects were various in different temperature levels. Two studies reported the effect of PM_{2.5} on the mortality was larger in low temperature (Li et al., 2013b; Sun et al., 2015), while one study found the effect was greater in high-temperature level and argued that cold air could be warmed in the nasal cavity and may not stimulate the respiratory system very much, so high temperature may be more important for respiratory health (Li et al., 2015). The mechanisms of the synergistic association between temperature and PM on mortality might be complex and are not yet clearly explained. Several explanations have been proposed for the synergistic effects of ambient particle pollution and high temperature on daily mortality. First, temperature plays an important role in determining patterns of air quality over multiple scales in time and space (Li et al., 2014; Macdonald et al., 2005). Second, high temperature has the positive effects on increased mortality (Barreca et al., 2015; Chen et al., 2015; Chung et al., 2015; Gao et al., 2015). Third, seasonal variations of the effect of PM₁₀ could partly be explained by variation in the chemical composition, in exposure, or by interaction with other parameters such as O₃ (Chen et al., 2012; Krall et al., 2013; Pascal et al., 2014). Besides, both temperature and PM have an obvious seasonal trend. Therefore, if high levels of PM and extreme temperature exist simultaneously, it could have a synergistic effect on human health.

3.5. Meta-analysis

16 studies from the 29 eligibility studies met the inclusion criteria of the meta-analysis and were chose to conduct the meta-analysis. The other 13 studies were excluded for three reasons: 6 were not focusing on PM₁₀ (Kioumourtzoglou et al., 2016; Li et al., 2013a, 2013b, 2015; Sun et al., 2015; Willers et al., 2016); Four were

not using stratification approach (Hales et al., 2000; Muggeo, 2007; Pinheiro et al., 2014; Willers et al., 2016) and three were not giving RR or β (Dholakia et al., 2014; Kim et al., 2015; Nawrot et al., 2007).

Tables 2–4 demonstrate the detailed information of studies included in the meta-analysis for low-temperature group, middle-temperature group and high-temperature group, respectively. Details of the meta-analysis results are showed in [supplementary material](#) (Figs. 3–5). The overall RRs of PM₁₀ on mortality in low temperature strata were 1.004 (95% CIs: 1.003 to 1.006), 1.005 (95% CIs: 1.003 to 1.007), and 1.005 (95% CIs: 1.000 to 1.010) for non-accidental death, cardiovascular death and respiratory death per 10 $\mu\text{g}/\text{m}^3$ increase in PM₁₀. In addition, the overall RRs of PM₁₀ on mortality in middle-temperature strata were 1.005 (95% CIs: 1.004 to 1.006), 1.005 (95% CIs: 1.004 to 1.007), and 1.008 (95% CIs: 1.006 to 1.010) for non-accidental death, cardiovascular death and respiratory death per 10 $\mu\text{g}/\text{m}^3$ increase in PM₁₀. Moreover, the overall RRs of PM₁₀ on mortality in high temperature strata were 1.012 (95% CIs: 1.010 to 1.015), 1.016 (95% CIs: 1.010 to 1.022) and 1.019 (95% CIs: 1.010 to 1.028) for non-accidental death, cardiovascular death and respiratory death per 10 $\mu\text{g}/\text{m}^3$ increase in PM₁₀. The present meta-analysis showed evidence that temperature modifies the effects of PM₁₀ on mortality. The PM₁₀ effect on mortality was greater on high-temperature days than on normal temperature days. The results of quantifying the differences of RRs of associations between PM₁₀ and mortality could be seen in the [supplementary material](#) (Table 2). In addition, the effects of PM₁₀ on mortality were the biggest in the high-temperature level, and the smallest in the low-temperature level. Similarly, Li et al. (2016) also found PM₁₀ effects on non-accidental mortality and cardiovascular mortality were greater on high-temperature days than on normal temperature days. Figs. 4–5 in the [supplementary material](#) show there was heterogeneity between studies. We use subgroup analyses to explore their heterogeneity. We roughly stratified the studies according to the duration time and geographic region. Specifically, four years was used as a cutoff point for the duration,

Table 2
Contextual details of studies on PM₁₀ in the low-temperature level included in the meta-analysis.

Author	Location	Period	Low-temperature level								
			Non-accidental death			Cardiovascular death			Respiratory death		
			β^a	95% CI		β^a	95% CI		β^a	95% CI	
Li et al., 2014	Guangzhou, China	2005–2009	0.960	–0.520	2.470	0.660	–1.660	3.030	1.970	–1.100	5.120
Shaposhnikov et al., 2014	Moscow, Russia	2006–2010	0.430	0.090	0.770	NA	NA	NA	NA	NA	NA
Pascal et al., 2014	French (9 cities)	2000–2006	0.800	0.100	1.400	0.800	0.100	1.500	–0.500	–1.900	0.900
Burkart et al., 2013	Berlin, Germany	1998–2010	0.200	–0.100	0.500	NA	NA	NA	NA	NA	NA
Burkart et al., 2013	Lison, Portugal	1998–2008	0.200	–0.500	0.800	NA	NA	NA	NA	NA	NA
Puza and Roberts, 2013	Melbourne, Australia	1995–2007	2.530	–9.863	18.400	NA	NA	NA	NA	NA	NA
Puza and Roberts, 2013	Sydney, Australia	1993–2007	0.774	–10.710	5.968	NA	NA	NA	NA	NA	NA
Cheng and Kan, 2012	Shanghai, China	2001–2004	0.460	0.210	0.720	0.490	0.130	0.860	0.240	–0.330	0.820
Meng et al., 2012	Guangzhou, China	2007–2008	2.800	1.930	3.660	3.180	1.800	4.550	3.050	1.110	5.000
Meng et al., 2012	Hangzhou, China	2002–2004	1.310	0.800	1.820	0.260	–0.640	1.160	1.770	0.690	2.850
Meng et al., 2012	Shanghai, China	2001–2008	0.190	–0.010	0.390	0.310	–0.010	0.630	–0.400	–0.950	0.140
Meng et al., 2012	Shenyang, China	2005–2008	–0.010	–0.310	0.300	0.170	–0.270	0.620	0.840	–0.110	1.780
Meng et al., 2012	Suzhou, China	2005–2008	0.170	–0.330	0.670	0.100	–0.680	0.890	–1.040	–2.380	0.030
Meng et al., 2012	Taiyuan, China	2004–2008	0.490	0.110	0.870	0.200	–0.430	0.830	1.450	0.220	2.680
Meng et al., 2012	Tianjin, China	2005–2008	1.560	0.750	2.360	0.950	–0.120	2.020	1.590	–1.410	4.600
Meng et al., 2012	Wuhan, China	2003–2005	0.370	–0.030	0.770	0.610	0.090	1.130	–0.910	–1.960	0.140
Li et al., 2011	Tianjin, China	2007–2009	0.280	0.080	0.490	0.250	–0.010	0.500	0.460	–0.120	1.040
Lin and Liao, 2009	Kaohsiung, China	1995–1999	–0.261	–1.727	1.240	–1.863	–4.866	1.110	NA	NA	NA
Qian et al., 2008	Wuhan, China	2001–2004	0.620	–0.090	1.340	0.720	–0.250	0.170	1.070	–0.760	2.950
Stafoggia et al., 2008	Italy (9 cities)	1997–2004	0.210	–0.060	0.470	0.310	–0.060	0.670	0.540	–0.470	1.570
Ren and Tong, 2006	Brisbane, Australia	1996–2001	0.151	–0.592	0.894	2.200	–2.880	7.540	NA	NA	NA
Roberts, 2004	Cook, Pennsylvania	1987–1994	–0.646	–2.140	0.850	NA	NA	NA	NA	NA	NA
Roberts, 2004	Allegheny, Pennsylvania	1987–1994	0.360	–2.660	3.480	NA	NA	NA	NA	NA	NA
Aga et al., 2003	28 European cities	Five years	0.440	0.250	0.640	NA	NA	NA	NA	NA	NA
Katsouyanni et al., 2001	29 European cities	Five years	0.290	0.160	0.420	NA	NA	NA	NA	NA	NA

^a Percent increment of mortality per 10 $\mu\text{g}/\text{m}^3$ increase in PM₁₀ in low temperature level.

Table 3Contextual details of studies on PM₁₀ in the middle-temperature level included in the meta-analysis.

Author	Location	Period	Middle-temperature level								
			Non-accidental death			Cardiovascular death			Respiratory death		
			β^a	95% CI		β^a	95% CI		β^a	95% CI	
Li et al., 2014	Guangzhou, China	2005–2009	0.600	0.320	0.890	0.650	0.170	1.140	1.350	0.750	1.950
Shaposhnikov et al., 2014	Moscow, Russia	2006–2010	0.770	0.400	1.130	NA	NA	NA	NA	NA	NA
Burkart et al., 2013	Berlin, Germany	1998–2010	0.800	0.200	1.300	NA	NA	NA	NA	NA	NA
Burkart et al., 2013	Lisbon, Portugal	1998–2008	0.040	–0.500	0.600	NA	NA	NA	NA	NA	NA
Puza and Roberts, 2013	Melbourne, Australia	1995–2007	0.296	–0.433	0.870	NA	NA	NA	NA	NA	NA
Puza and Roberts, 2013	Sydney, Australia	1993–2007	0.735	0.022	1.843	NA	NA	NA	NA	NA	NA
Cheng and Kan, 2012	Shanghai, China	2001–2004	0.200	0.060	0.340	0.260	0.060	0.470	0.260	–0.070	0.600
Meng et al., 2012	Guangzhou, China	2007–2008	1.080	0.850	1.310	1.250	0.880	1.620	1.410	0.890	1.940
Meng et al., 2012	Hangzhou, China	2002–2004	0.420	0.140	0.690	0.100	–0.370	0.570	0.760	0.160	1.370
Meng et al., 2012	Shanghai, China	2001–2008	0.410	0.320	0.500	0.440	0.300	0.580	0.690	0.460	0.930
Meng et al., 2012	Shenyang, China	2005–2008	0.330	0.160	0.490	0.390	0.150	0.630	0.780	0.270	1.300
Meng et al., 2012	Suzhou, China	2005–2008	0.430	0.220	0.640	0.390	0.050	0.730	0.550	–0.020	1.110
Meng et al., 2012	Taiyuan, China	2004–2008	0.360	0.180	0.530	0.310	0.020	0.600	0.950	0.340	1.550
Meng et al., 2012	Tianjin, China	2005–2008	0.780	0.470	1.080	0.980	0.580	1.370	0.570	–0.700	1.850
Meng et al., 2012	Wuhan, China	2003–2005	0.620	0.460	0.780	0.690	0.480	0.910	0.850	0.410	1.290
Park et al., 2011	Seoul, Korea	1999–2007	0.420	0.030	0.810	NA	NA	NA	NA	NA	NA
Lin and Liao, 2009	Kaohsiung, China	1995–1999	0.260	–0.877	1.410	–1.500	–2.457	2.119	NA	NA	NA
Qian et al., 2008	Wuhan, China	2001–2004	0.360	0.170	0.560	0.390	0.110	0.660	0.800	0.250	1.350
Stafoggia et al., 2008	Italy (9 cities)	1997–2004	1.600	0.640	2.570	2.050	0.470	3.660	3.150	0.640	5.730
Roberts, 2004	Cook, USA	1987–1994	0.686	0.330	1.036	NA	NA	NA	NA	NA	NA
Roberts, 2004	Allegheny, USA	1987–1994	0.567	0.080	1.065	NA	NA	NA	NA	NA	NA

^a Percent increment of mortality per 10 $\mu\text{g}/\text{m}^3$ increase in PM₁₀ in middle temperature level.**Table 4**Contextual details of studies on PM₁₀ in the high-temperature level included in the meta-analysis.

Author	Location	Period	High-temperature level								
			Non-accidental death			Cardiovascular death			Respiratory death		
			β^a	95% CI		β^a	95% CI		β^a	95% CI	
Shaposhnikov et al., 2014	Moscow, Russia	2006–2010	1.440	0.940	1.940	NA	NA	NA	NA	NA	NA
Li et al., 2014	Guangzhou, China	2005–2009	2.340	0.550	4.160	1.350	–1.690	4.480	6.090	2.420	9.890
Pascal et al., 2014	French (9 cities)	2000–2006	2.200	–1.400	5.800	2.300	–5.500	10.700	–3.500	–13.800	8.000
Burkart et al., 2013	Berlin, Germany	1998–2010	3.100	2.400	3.900	NA	NA	NA	NA	NA	NA
Burkart et al., 2013	Lisbon, Portugal	1998–2008	1.240	1.900	2.900	NA	NA	NA	NA	NA	NA
Puza and Roberts, 2013	Melbourne, Australia	1995–2007	0.095	–3.196	2.166	NA	NA	NA	NA	NA	NA
Puza and Roberts, 2013	Sydney, Australia	1993–2007	0.962	–2.172	6.192	NA	NA	NA	NA	NA	NA
Cheng and Kan, 2012	Shanghai, China	2001–2004	0.440	0.070	0.820	0.720	0.160	1.290	NA	NA	NA
Meng et al., 2012	Guangzhou, China	2007–2008	2.500	1.870	3.120	3.090	2.050	4.140	3.210	1.800	4.610
Meng et al., 2012	Hangzhou, China	2002–2004	0.450	–0.320	1.210	0.060	–1.280	1.390	1.840	0.130	3.540
Meng et al., 2012	Shanghai, China	2001–2008	0.870	0.610	1.120	1.230	0.810	1.650	0.370	–0.420	1.160
Meng et al., 2012	Shenyang, China	2005–2008	0.440	0.190	0.690	0.150	–0.230	0.520	1.090	0.290	1.890
Meng et al., 2012	Suzhou, China	2005–2008	2.170	1.610	2.720	2.700	1.770	3.630	1.230	–0.370	2.820
Meng et al., 2012	Taiyuan, China	2004–2008	0.830	0.450	1.220	0.440	–0.220	1.090	0.340	–1.210	1.890
Meng et al., 2012	Tianjin, China	2005–2008	1.230	0.420	2.030	1.790	0.740	2.850	2.530	–0.670	5.740
Meng et al., 2012	Wuhan, China	2003–2005	2.420	1.930	2.910	3.190	2.550	3.840	4.350	3.020	5.690
Li et al., 2011	Tianjin, China	2007–2009	0.620	0.270	0.970	0.920	0.470	1.360	0.740	–0.330	1.820
Park et al., 2011	Seoul, Korea	1999–2007	0.630	0.250	1.010	NA	NA	NA	NA	NA	NA
Lin and Liao, 2009	Kaohsiung, China	1995–1999	0.518	–0.833	1.910	0.775	–2.045	3.583	NA	NA	NA
Qian et al., 2008	Wuhan, China	2001–2004	2.200	0.740	3.680	3.280	1.240	5.370	1.150	–3.540	6.070
Stafoggia et al., 2008	Italy (9 cities)	1997–2004	2.550	1.580	3.520	2.680	1.200	4.170	4.120	0.440	7.930
Ren and Tong, 2006	Brisbane, Australia	1996–2001	4.180	1.010	7.460	6.220	0.930	11.780	NA	NA	NA
Roberts, 2004	Cook, USA	1987–1994	0.952	0.530	1.370	NA	NA	NA	NA	NA	NA
Roberts, 2004	Allegheny, USA	1987–1994	0.640	0.040	1.245	NA	NA	NA	NA	NA	NA
Aga et al., 2003	28 European cities	Five years	0.910	0.770	1.050	NA	NA	NA	NA	NA	NA
Katsouyanni et al., 2001	29 European cities	Five years	0.820	0.690	0.960	NA	NA	NA	NA	NA	NA

^a Percent increment of mortality per 10 $\mu\text{g}/\text{m}^3$ increase in PM₁₀ in high temperature level.

and study sites were grouped into Asia area, Europe area, America area and Oceania area. By subgroup analyses, we found that the duration period and continent division could be the sources of heterogeneity. Table 3 and Figs. 6–10 in the supplementary material show I^2 and the forest plots for subgroup analyses. The results of Egger's tests did not suggest obvious evidence of publication bias. In addition, the sensitive analyses showed that the result was robust, and the results could be seen in the

supplementary material (Table 4).

3.6. Modeling strategy for interaction

There are three main ways of the 29 studies to explore the modification effects, which are product terms, stratification analysis and bivariate response surface, and the same as the description of the two studies (Ren and Tong, 2006; Roberts, 2004). The

product term approach involves a model that includes a multiplicative interaction term between air pollutants and meteorological factors, and the two factors are equally treated, which is most commonly used for interactions in statistical modeling (Puza and Roberts, 2013). The stratification approach involves stratify original data by one factor which was as the modifier, like the meteorological variable, to different datasets, and then fitting models separately allowing the effects of air pollutants to differ between these strata and vice versa. The bivariate response surface approach involves modeling the relationship between air pollutants and meteorological factors as a surface often using nonparametric models. These three approaches have their own corresponding advantages and disadvantages. Particularly, the bivariate response surface approach does not contain parametric assumptions and is easy to fit using existing soft wares with a complete description of the interaction effects, but can be hard to get explicitly quantify the effects of air pollutants or meteorological factors (Morris and Naumova, 1998). Similarly, the product term approach is also easy to fit, but it is hard to interpret. Additionally, the stratification approach is the most easily interpretable in the three approaches with fewer parameters and giving a simple, quantitative comparison of the effect of meteorological factors in the different air pollutants strata and vice versa. However, stratification approach needs the specification of cut points that determine the strata, and there is no uniform standard method to get the cut points. The factors' cut-off points are arbitrary. In the eligible studies, most of them adopted different percentiles or mean of the factors as the cut-points, or based on statistical and biological considerations to choose different levels of temperature. In addition, one study considered the temperature cut-off points as the one parameter to estimate in the Bayesian model, and used Markov chain Monte Carlo methods to perform the inference (Puza and Roberts, 2013). Overall, it is suggested to use the first form of product term analysis to help acquire the possible interaction effects and use stratification approach or the other two forms of product term approach to get more precise interaction effects. Meanwhile, the bivariate response surface can be used to investigate whether an interaction seems plausible and as a guide in getting the location of the cut points and selecting appropriate strata for use in a stratification analysis or the product term models.

3.7. Future research direction

Focusing on the inactive effect between other meteorological factors and ambient air pollutants on other health outcomes, like years of life lost (YLL), is an important research direction. Further epidemiology evidence from more counties or cities with different climatic conditions and topographic conditions are needed to get a more comprehensive knowledge of the interaction effects. Moreover, when using stratification or product terms to model the interaction, there should be a standard method to select cut-offs of air pollutants and meteorological factors.

3.8. Limitations

First, the 29 eligible studies in this review are observational study, so the strength of the evidence is not strong. However, the most commonly used study type in assessing the effects of air pollution on mortality was observational study. Second, In the meta-analysis, temperatures were grouped into three levels: "low", "middle" and "high", whose definitions of low, middle and high level may be not consistent across the studies. Most the original studies classified temperature into different levels depending on the percentile of temperature. However, we use the study-specific cut points for temperature and assume different people in

different place have their own optimal temperatures for health.

4. Conclusions

Overall, the temperature modifies the effect of PM₁₀ on mortality. The effect of PM₁₀ on mortality is greatly enhanced in the high temperature, especially on respiratory death. Our results highlight the importance of considering the interaction between PM₁₀ and temperature on mortality in future research. Also, policy makers should enhance the emission control of PM₁₀ in high-temperature days.

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Abbreviations

PM	(particulate matter)
PM _{2.5}	(particulate matter with less than 2.5 µg/m ³ in aerodynamic diameter)
PM ₁₀	(particulate matter with less than 10 µg/m ³ in aerodynamic diameter)
O ₃	(ozone)
YLL	(Years of Life Lost)
GLM	(Generalized linear model)
GAM	(Generalized additive model)
DLNM	(Distributed Lag Linear and Non-Linear Model)
DLM	(Distributed Lag Linear Model)
BRS	(Bivariate Response Surfaces)
COPD	(Chronic obstructive pulmonary disease)

Appendix A. Supplementary data

Supplementary data related to this article can be found at <http://dx.doi.org/10.1016/j.envpol.2017.02.012>.

Financial interests

No competing financial interests declaration.

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